MOCK TEST 67

📖 READING

5 parts · 35 questions

⏱ 60 minutes

Vaqt: 60 daqiqa. Javoblarni javoblar varaqasiga o'tkazishni unutmang.

PART 1

Questions 1–6

Read the text. Fill in each gap with ONE word. You must use a word which is somewhere in the rest of the text.

COLOUR ME HEALTHY

Different Colours can affect us in many different ways; that's according to Verity Allen. In her new series 'Colour me Healthy', Verity looks at the ways that **Q1**_____ can influence how hard we work and the choices we make. They can even change our emotions and even influence how healthy we are. 'Have you ever noticed how people always use the same colours for the same things?' says Verity. 'Our toothpaste is always white or blue or maybe red. It's never green. Why not? For some reason we think that blue and white is clean, while we think of green products as being a bit disgusting. We respect a company which writes its name in blue or black, but we don't respect one that uses pink or orange. **Q2**_____ who design new **Q3** _____ can use these ideas to influence what we buy.' During this four-part **Q4**_____ Verity studies eight different colours, two colours in each programme. She meets people who work in all aspects of the colour industry, from people who **Q5**_____ food packets, to people who name the colours of lipsticks. Some of the people she meets clearly have very little scientific knowledge to support their **Q6** _____, such as the American 'Colour Doctor' who believes that serious diseases can be cured by the use of coloured lights. However, she also interviews real scientists who are studying the effects of green and red lights on mice, with some surprising results. Overall, it's an interesting show, and anyone who watches it will probably find out something new. But because Verity goes out of her way to be polite to everyone she meets on the series, it's up to the viewers to make their own decisions about how much they should believe.

PART 2

Questions 7–14

There are descriptions of eleven holidays. Decide which holiday (letters A to K) would be the most suitable for the people described below.

7

You are a very busy investment banker who wants a week off work. You like scenery and fresh air. You are very fit and you enjoy walking. For this holiday you want to leave the beach and try something different.

8

You are a student. You will go to university next year, and your parents want to give you a holiday as a present before you go. You like reading and dancing, and want somewhere romantic and relaxing

9

You want to give your husband a holiday to celebrate his 60th birthday. You like all sorts of sport, but because your heart is rather weak, you cannot do anything too energetic. You would like to stay in a good hotel.

10

You own your own business. You have had a successful year, and you want a holiday. You want somewhere with very good food and accommodation, and you want to enjoy yourself without doing very much.

11

The Hamble family are looking for something that will interest Mark aged 7 and Tammy, aged 13. They all like animals and sport. Mr and Mrs Hamble would also like something that lets them do some exploring on their own.

12

Sammie wants to visit a market after spending the day in the city. He would like to photograph a historic place, and buy a painting by someone unknown.

13

Alexia is looking for a really special necklace for her grandmother's birthday. She'd like to spend the whole day at the market, and wants to avoid the cold by staying inside.

14

Ella is looking for objects from other countries for her friends. She'd like to choose a second-hand book to read on the journey home, and wants a snack at the market, too.

A) Golf in America: A golfing safari across the United States in just ten days. Try eight great courses from New England to California, with golf lessons as you go. Transport is provided between your 5 star hotels and golf courses.

B) Climbing in Switzerland: Whether you are experienced or just beginning, Hipeak have the mountaineering holiday for you. Choose one of our seven, ten or fourteen day packages, and you can enjoy wonderful alpine scenery while climbing at your particular level of ability. No mountaineering gear is needed - we supply it all.

C) Florida and Disney world: Every kid dreams of Disney World, and after mom and dad have seen it all, you can leave the children with one of our special guides, and try some scuba diving in the Florida Keys, and a shopping tour of Miami. Or you can see Florida for yourselves with our no-problem car-hire scheme.

D) Safari in Kenya: Try seven days in Wariwaro lodge in Kenya's Serengetti park. Coach tours through the savannah to see some of the world's most spectacular wildlife. Flights go every Sunday from Heathrow, London, and there is our special family option, with children getting a one-third discount.

E) Holiday in Majorca: Two wild weeks in Majorca. Yes, if you are aged between 18 and 31, this package holiday is for you. Thr fun starts at Manchester airport, and in Majorca the party moves from the beach to the disco to the swimming pool. It only stops when you can't party any more. If you want a challenge, try this!

F) Mediterranean cruise: The ancient Mediterranean in 16 days. Take a luxury cruise on the RMV Perikles to Herculaneum, Caesarea, Athens and Alexandria. Enjoy our on-board facilities, including restaurant, library, cinema and disco.

G) Carribean Beach: You've been working too hard! Forget it all on a tropical beach. Ten days in a luxury hotel, right on the beach. You hardly have to move from your deckchair, except to enjoy our award-winning cookery. Beach cricket and barbeques are on offer for the more energetic.

H) Hiking in Katmandu: Try a Katmandu adventure. See the Himalayas for yourself in this two week hiking tour. You must be fit and an experienced hiker, as well as having your own boots. We supply tents, the guides and the holiday of a lifetime!

I) Rosewell Hill: Our market's in an amazing building that's hundreds of years old. Visitors find our late-night opening hours convenient, and there are always performers entertaining the crowds. We've recently opened more stalls specializing in pictures both from wellknown artists and also those beginning their careers.

J) Camberwall Market: There's lots to see in this interesting indoor market, so it's open from morning until late, in a fantastic modern setting. Find everything from rare gold and silver jewellery to designer clothes - although the prices aren't cheap, the quality's excellent. After shopping, enjoy a meal in a nearby restaurant.

K) Frome Place: Stalls open during normal daytime shopping hours so, depending on the weather, there's plenty to entertain you the whole day. Try our sandwich bar if you're hungry, and look for an old copy of something by a favourite author. We also have gifts from all over the world.

PART 3

Questions 15–20

Read the text and choose the correct heading for each paragraph from the list of headings below. There are more headings than paragraphs, so you will not use all of them. You cannot use any heading more than once.

List of Headings:

A) British chemists' achievement (example)
B) The creation of six-membered rings
C) Products produced by Diels-Alder reactions
D) The large receptor molecule designed by British chemists
E) The drawback of the receptor
F) The Diels-Alder reaction
G) Fast reactions due to the receptor
H) Further efforts to be made on the receptor

15. Paragraph I **16. Paragraph II** **17. Paragraph III**
18. Paragraph IV **19. Paragraph V** **20. Paragraph VI**

DIELS-ALDER RECEPTOR MOLECULE

I. Jeremy Sanders and his colleagues at the University of Cambridge have designed and synthesized a large cyclic "receptor" molecule, which makes one such reaction proceed almost 60,000 times as fast as usual. The receptor is similar to another built last year by the same team (New Scientist, Science, 1 February 1992). It consists of a ring of three porphyrin molecules linked by bridging chemical groups. Each porphyrin molecule contains a zincion at its centre. The central cavity of the new receptor is slightly smaller than before, and the researchers have also anchored pyridine groups to two of the zincions to act as bonding sites.

II. Sanders and his colleagues have used their receptor to speed up and control the products of the so-called Diels-Alder reaction, a mainstay of chemical synthesis. The reaction occurs between two molecules - a "diene", which has two carbon-carbon bonds separated by a single bond, and a diene-seeking molecule, or "dienophile". In the right conditions, these two molecules transfer the electrons involved in their double bonds from atom to atom to form new bonds that complete a ring of six carbon atoms with a single double bond. The creation of such six-membered rings is the crucial first step in making many pharmaceuticals and agrochemical.

III. Some Diels-Alder reactions are too slow to be useful industrially. The researchers, therefore, designed their receptor so that it would hold the diene and dienophile, via the pyridine (Py) groups, in the right positions to react quickly. According to Sanders, the receptor acts like a "molecular reaction vessel in which the effective concentration of reactants can be increased dramatically, so allowing a fast reaction".

IV. Normally, Diels-Alder reactions produce a mixture of two products. But because in the receptor the reactants are forced into a specific orientation relative to each other, only one of the two possible products can form.

V. Sanders hopes to modify the receptor to bring together in the cavity two molecules that do not normally react. This could lead to be the synthesis of compounds which everyday synthetic chemistry cannot make.

VI. The receptor differs from an enzyme or other catalyst in one important respect. Only a tiny amount of an enzyme is needed

to make a reaction thousands of times faster, but large quantities of the receptor are needed to make a significant difference to the speed of a reaction. However, Sanders is confident that in the future his team will be able to increase the turnover or able to increase the turnover of reactants by designing new features into the receptor. This would reduce the amount of receptor needed to speed up a reaction by a given amount. The researchers report further details of their results in the latest issue of Journal of the Chemical Society, Chemical Communications.

PART 4

Questions 21–29

Read the following text for questions 21-29.

HOW BACTERIA INVENTED GENE EDITING

This week the UK Human Fertilisation and Embryology Authority okayed a proposal to modify human embryos through gene editing. The research, which will be carried out at the Francis Crick Institute in London, should improve our understanding of human development. It will also undoubtedly attract controversy - particularly with claims that manipulating embryonic genomes is a first step towards designer babies. Those concerns shouldn't be ignored. After all, gene editing of the kind that will soon be undertaken at the Francis Crick Institute doesn't occur naturally in humans or other animals.

It is, however, a lot more common in nature than you might think, and it's been going on for a surprisingly long time - revelations that have challenged what biologists thought they knew about the way evolution works. We're talking here about one particular gene editing technique called CRISPR-Cas, or just CRISPR. It's relatively fast, cheap and easy to edit genes with CRISPR - factors that explain why the technique has exploded in popularity in the last few years. But CRISPR wasn't dreamed up from scratch in a laboratory. This gene editing tool actually evolved in single-celled microbes.

CRISPR went unnoticed by biologists for decades. It was only at the tail end of the 1980s that researchers studying Escherichia coli noticed that there were some odd repetitive sequences at the end of one of the bacterial genes. Later, these sequences would be named Clustered Regularly Interspaced Short Palindromic Repeats - CRISPRs. For several years the significance of these CRISPRs was a mystery, even when researchers noticed that they were always separated from one another by equally odd 'spacer' gene sequences.

Then, a little over a decade ago, scientists made an important discovery. Those 'spacer' sequences look odd because they aren't bacterial in origin. Many are actually snippets of DNA from viruses that are known to attack bacteria. In 2005, three research groups independently reached the same conclusion: CRISPR and its associated genetic sequences were acting as a bacterial immune system. In simple terms, this is how it works. A bacterial cell generates special proteins from genes associated with the CRISPR repeats (these are called CRISPR associated - Cas - proteins). If a virus invades the cell, these Cas proteins bind to the viral DNA and help cut out a chunk. Then, that chunk of viral DNA gets carried back to the bacterial cell's genome where it is inserted - becoming a spacer. From now on, the bacterial cell can use the spacer to recognise that particular virus and attack it more effectively.

These findings were a revelation. Geneticists quickly realised that the CRISPR system effectively involves microbes deliberately editing their own genomes - suggesting the system could form the basis of a brand new type of genetic engineering technology. They worked out the mechanics of the CRISPR system and got it working in their lab experiments. It was a breakthrough that paved the way for this week's announcement by the HFEA. Exactly who took the key steps to turn CRISPR into a useful genetic tool is, however, the subject of a huge controversy. Perhaps that's inevitable - credit for developing CRISPR gene editing will probably guarantee both scientific fame and financial wealth.

Beyond these very important practical applications, though, there's another CRISPR story. It's the account of how the discovery of CRISPR has influenced evolutionary biology. Sometimes overlooked is the fact that it wasn't just geneticists who were excited by CRISPR's discovery - so too were biologists. They realised CRISPR was evidence of a completely unexpected parallel between the way humans and bacteria fight infections. We've known for a long time that part of our immune system "learns" about the pathogens it has seen before so it can adapt and fight infections better in future. Vertebrate animals were thought to be the only organisms with such a sophisticated adaptive immune system. In light of the discovery of CRISPR, it seemed some bacteria had their own version. In fact, it turned out that lots of bacteria have their own version. At the last count, the CRISPR adaptive immune system was estimated to be present in about 40% of bacteria. Among the other major group of single- celled microbes - the archaea - CRISPR is even more common. It's seen in about 90% of them. If it's that common today, CRISPR must have a history stretching back over millions - possibly even billions - of years. "It's clearly been around for a while," says Darren Griffin at the University of Kent.

The animal adaptive immune system, then, isn't nearly as unique as we thought. And there's one feature of CRISPR that makes it arguably even better than our adaptive immune system: CRISPR is heritable. When we are infected by a pathogen, our adaptive immune system learns from the experience, making our next encounter with that pathogen less of an ordeal. This is why vaccination is so effective: it involves priming us with a weakened version of a pathogen to train our adaptive immune system. Your children, though, won't benefit from the wealth of experience locked away in your adaptive immune system. They have to experience an infection - or be vaccinated - first hand before they can learn to deal with a given pathogen.

Do the following statements agree with the given information?

21 **The research carried out at the Francis Crick Institute in London is likely to be controversial.**

A) True
B) False
C) Not Given

22 **Gene editing, like the one in the upcoming research, can happen naturally in humans or other animals.**

A) True
B) False
C) Not Given

23 **CRISPR-Cas is a gene editing technique.**

A) True
B) False
C) Not Given

24 **CRISPR was noticed when the researchers saw some odd repetitive sequences at the ends of all bacterial genes.**

A) True
B) False
C) Not Given

25 **A group of American researchers made an important revelation about the CRISPR.**

A) True
B) False
C) Not Given

Choose the correct letter, A, B, C or D.

26 **'Spacer' sequences look odd because:**

A) they are a bacterial immune system
B) they are DNA from viruses
C) they aren't bacterial in origin
D) all of the above

27 **The ones, who were excited about the CRISPR's discovery, were:**

A) biologists
B) geneticists
C) physicists
D) A and B

28 **Word "learns" in the 6th paragraph means:**

A) determines
B) gains awareness
C) adapts
D) studies

29 **What makes CRISPR better than even our adaptive immune system?**

A) long history of existence
B) immortality
C) heritability
D) adaptiveness

PART 5

Questions 30–35

Read the following text for questions 30-35.

THE GIG ECONOMY

The gig economy is a labor market characterized by the prevalence of short- term contracts or freelance work as opposed to permanent jobs. This type of economy has grown rapidly with the advent of digital platforms that connect freelancers with clients. Companies like Uber, TaskRabbit, and Upwork have revolutionized the way people work, offering flexibility and new opportunities for earning income.

One of the primary advantages of the gig economy is flexibility. Workers can choose when, where, and how much they work. This is particularly beneficial for individuals who need to balance work with other commitments, such as students, caregivers, or those pursuing creative projects. Additionally, the gig economy allows workers to diversify their income streams, reducing dependence on a single employer and potentially increasing overall earnings.

However, the gig economy also presents significant challenges. Job security is a major concern as gig workers are often classified as independent contractors rather than employees., This classification means they typically do not receive paid leave, or retirement contributions. Furthermore, income in the gig economy can be unpredictable and inconsistent, benefits such as health insurance, making financial planning difficult.

The legal status of gig workers has been a contentious issue. Many argue that gig workers should be entitled to the same protections and benefits as traditional employees. Some countries and states have begun to address this issue. For example, California passed Assembly Bill 5 (AB5), which aimed to reclassify many gig workers as employees. While the law has faced challenges and pushback, it represents a significant step toward redefining the rights of gig workers.

The gig economy also impacts the traditional job market. On one hand, it creates new opportunities for employment and entrepreneurship. On the other hand, it can lead to increased competition and downward pressure on wages. Companies benefit from lower labor costs and increased flexibility, but this can come at the expense of workers' rights and job stability.

Another critical aspect of the gig economy is the role of technology. Digital platforms facilitate the connection between workers and clients, making it easier to find and complete gigs. However, these platforms often charge fees and commissions, which can reduce the overall earnings of gig workers. Additionally, there is a growing concern about the data privacy and security of gig workers, as these platforms collect extensive personal information.

The COVID-19 pandemic has further highlighted the vulnerabilities and potential of the gig economy. During the pandemic, many people turned to gig work as traditional jobs were lost or furloughed. Gig workers played a crucial role in delivering essential services, such as food delivery and transportation. However, the lack of benefits and protections for gig workers became more apparent, prompting calls for reform and better support for this workforce.

Looking forward, the gig economy is expected to continue growing, driven by technological advancements and changing work preferences. However, it will be essential to address the challenges associated with this type of work. Policies and regulations need to evolve to ensure that gig workers receive fair treatment and adequate protections.

Fill in the missing information. Write no more than ONE WORD and/or A NUMBER for each question.

The gig economy offers significant **30** _____ in terms of when and where individuals can work. Gig workers typically do not receive benefits such as health insurance or **31** _____ leave.

The classification of gig workers has been a **32** _____ issue. Digital platforms often charge **33** _____ and commissions, which can reduce gig workers' earnings.

Choose the correct answer A, B, C, or D.

34 **What is a significant challenge faced by gig workers due to their classification as independent contractors?**

A) Lack of flexibility
B) Unpredictable work hours
C) Absence of benefits
D) Increased job security

35 **How did the COVID-19 pandemic highlight the vulnerabilities of gig workers?**

A) By reducing the availability of gig work
B) By increasing the demand for traditional jobs
C) By emphasizing the lack of benefits and protections
D) By providing more stable income opportunities

MOCK TEST 67

35 answers

1 colours / colors
2 people
3 products
4 series
5 design
6 ideas
7 B
8 F
9 A
10 G
11 C
12 I
13 J
14 K
15 D
16 F
17 G
18 C
19 H
20 E
21 True
22 False
23 True
24 False
25 Not Given
26 C / they aren't bacterial in origin
27 D / A and B
28 B / gains awareness
29 C / heritability
30 flexibility
31 paid
32 contentious
33 fees
34 C / Absence of benefits
35 C / By emphasizing the lack of benefits and protections